



strategies, allowing designers to fine-tune the vehicle's architecture for optimal performance.

Building a versatile development platform for vehicle design is another focus area. This platform should be capable of handling multiple design variations, including concept vehicles, and adaptable to new designs and optimisations. For instance, a leading automotive company has successfully used such a platform to reduce the weight of their vehicles and manage over 300 different variations.

These advancements in aerodynamics, vehicle integration, and PCB design are crucial in shaping the future of EV development, making them safer, more efficient, and more adaptable to a range of applications.

Integration of embedded hardware and software

A crucial component in the development of EVs is the integration of embedded hardware and software. This aspect goes beyond the realms of MCAD and electronic CAD (ECAD), encompassing a vast array of software components that have crucial functions at various stages of vehicle development.

The complexity of modern EVs necessitates a high level of simulation and software management. In most organisations, different sets of software are employed at each development stage. This includes model-based controls, software controls, and multi-domain, multi-language software, all of which need to be effectively synchronised.

To manage this complexity, the concept of application lifecycle management (ALM) has become increasingly important. While data management in the context of CAD, computer-aided engineering (CAE), manufacturing, and shop floor data is familiar, ALM extends this concept to track, store, and manage

the life cycle of applications. This is particularly relevant for EVs, where each new model or software update requires meticulous documentation and tracking.

ALM allows for a comprehensive view of the entire development process, from embedded hardware and software to vehicle integration. This holistic approach is crucial in managing the complexities of modern EVs, ensuring that all components are not only individually optimised but also seamlessly integrated. Advanced ALM tools like Polarion facilitate this process, enabling organisations to handle various aspects of EV development within a single, unified environment.

Keeping track of every update, along with its history and integration with the vehicle's 3D models, is vital for maintaining the integrity and functionality of EVs. The use of ALM in the automotive industry represents a significant step forward in the efficient and effective development of EVs, ensuring that they are not only technologically advanced but also reliable and safe for consumers.

The final pillar

The final critical pillar in vehicle electrification is energy management, which involves a comprehensive analysis of energy consumption throughout the vehicle's operation. Key factors include the impact of thermal packaging on energy efficiency and the role aerodynamics plays in optimising energy usage.

Testing these various aspects of energy management is a two-pronged approach. Initially, much of the testing and analysis is conducted virtually using sophisticated software. However, physical testing remains an essential part of the process to ensure real-world applicability and effectiveness.

In the realm of noise, vibration,

and harshness (NVH) in electric and hybrid vehicles, specialised solutions are employed. These solutions include hardware setups that enable the analysis and optimisation of NVH performance. This is crucial, as the quieter operation of EVs brings NVH characteristics to the forefront.

The integration of test results with CAE methods leads to the development of new models and techniques, which are in turn utilised in manufacturing processes. For instance, comprehensive testing solutions that can bridge the gap between virtual simulations and physical testing are available in the market. These systems allow for seamless integration with both the virtual models and actual hardware used in vehicles.

One of the case studies in this field involves a leading automotive manufacturer that has effectively utilised these testing solutions. They have managed to create flexible development platforms that support future projects, showcasing the versatility and adaptability of these energy management systems.

As the automotive industry transitions towards EVs, it confronts a range of challenges, from battery production intricacies to integrating advanced electronics. By embracing cutting-edge simulation techniques, innovative materials, and sophisticated design tools, the industry is not only adapting to these changes but also shaping the future of sustainable transportation. These advancements, grounded in technology and driven by efficiency and safety, position the automotive sector at the forefront of a greener, electric-powered future. **EFY**

This article is based on a tech talk session at EFY Expo Pune 2023 by Prabodhan Purvant, CTO, and Ashish Gawade, Global Sales Head, 3D Engineering. It has been transcribed and curated by Nidhi Agarwal, Technology Journalist at EFY.